Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY **9700/23**

Paper 2 AS Level Structured Questions **May/June 2020**

MARK SCHEME

Maximum Mark: 60

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

© UCLES 2020

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

© UCLES 2020

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2020

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided • Any response marked *ignore* in the mark scheme should not count towards ***n*** • Incorrect responses should not be awarded credit but will still count towards ***n*** • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

© UCLES 2020

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

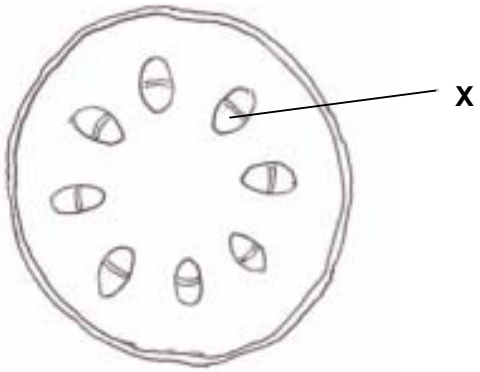
Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations:

;	separates marking points
/	alternative answers for the same marking point
R	reject
A	accept
I	ignore
AVP	any valid point
AW	alternative wording (where responses vary more than usual)
ecf	error carried forward
underline	actual word underlined must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument

© UCLES 2020

Question	Answer	Marks
1(a)	any one correct location **;** **R** if more than one label line and one is to an incorrect location *allow the mark if more than one label line and <u>all</u> point to xylem locations*  	**1**
1(b)	**I** *ref. to* adhesion / H bonding between water molecules and cellulose lining *any three from* (hydrogen bonding causes) <u>cohesion</u> between water molecules **;** water molecules attracted to each other /AW **;** water leaving xylem (at top), pulls water molecules below **;** transpiration pull **;** *idea that* contributes to, an unbroken / a continuous, column of water (within xylem vessels) **;** AVP **;** e.g. allows movement of water against (pull of) gravity	**3**

Question	Answer	Marks
1(c)(i)	*any three from* *description max 2* (movement to adjacent cell) via plasmodesmata **;** (movement) through symplasm / cytoplasm / symplastic pathway / cytoplasmic pathway **;** (includes) movement through vacuole / vacuolar pathway **;** *explanation* cell **B** has a lower water potential than **A** **ora** *or* cell **C** has a lower water potential than **B** **; ora** *or* water moves down the water potential gradient / from high to low water potential / to lower water potential / from less negative to more negative water potential **;** **R** from high to low water potential gradient **R** to lower water potential gradient water moves, into / out of, vacuole by osmosis *or* water crosses tonoplast by osmosis **;**	**3**
1(c)(ii)	*any three from* cell wall **;** large (permanent) vacuole **;** tonoplast **;** plasmodesmata / cytoplasmic strands between adjacent cells **;** nucleus at, edge / periphery, of cell / AW **;** AVP **;** many small Golgi bodies **A** dictyosomes starch, granules / grains **I** lack of centrioles **I** chloroplasts	**3**

Question	Answer	Marks
2(a)	*allow mosquito for Anopheles throughout* *allow pathogen for Plasmodium throughout* *any three from* role of *Anopheles* in transmission cycle **;** e.g. *Anopheles* is, a vector of *Plasmodium* / *Anopheles* passes *Plasmodium* from infected person to uninfected person *insecticide on nets and on surfaces* kills *Anopheles* before it can take blood from an infected person **;** kills *Anopheles* before it can transfer blood with *Plasmodium* to uninfected person **;** *presence of nets* protect people, when sleeping / at time when *Anopheles* is, active / feeding **;** *general* prevent *Plasmodium* from completing its life cycle **;** AW AVP **;** *idea of* reducing population size of mosquitoes use of different insecticides on net and IRS to avoid insecticide resistance	**3**
2(b)(i)	*any one from* increase in the use of ITN over time **;** decrease in the use of IRS only **;** proportion of population protected by ITN only has increased **;**	**1**
2(b)(ii)	*any one from* increase is not steep enough, to make a large enough difference / so disease will still be able to spread **;** a large proportion of the population is still at risk, so people will still contract the disease **; A** figures from Fig. 2.1 WHO targets may not be met, so hindering progress in the fight against malaria **;** AW AVP **;** e.g. suggests that the, provision of / distribution of / access to, ITN is not adequate **;**	**1**

© UCLES 2020

Question	Answer	Marks
2(b)(iii)	*any one suggestion from* ITN more effective in control than IRS **;** more cost effective to provide ITN **;** lack of personnel to carry out work required for IRS **;** *ref. to* insecticide on net may be more effective at killing ; AVP **;** e.g. outside agencies / AW, provide ITN but do not help with IRS	**1**
2(c)	*any three from* higher concentration of antibodies **;** faster production of antibodies **;** because of presence of memory, B-lymphocytes / B cells **;** higher numbers of specific B-lymphocytes, so increased chance of faster recognition / because of clonal expansion in first response **;** AVP **;** e.g. also more memory T-cells to stimulate B-lymphocyte response *ref. to* higher concentration antibodies in circulation remaining after recovery	**3**
2(d)	*any two from* result of an autoimmune disease / AW **;** antibodies produced against, self-antigens / antigens on body cells *or* antibodies bind to self-antigens / antigens on (own) body cells **;** detail **;** e.g. prevents functioning of muscle cells binds to receptors on muscle cells	**2**

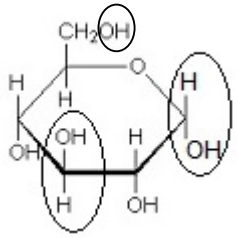
Question	Answer	Marks
3(a)(i)	**X** = artery **Y** = vein **;**	**1**
3(a)(ii)	*any two from* cross section not regular / no defined shape / AW **;** **A** not circular tunica intima smooth**;** **A** inner layer *for tunica intima* **A** not, crinkly / wavy thin / thinner (than **X**) tunica media **;** **A** thin middle layer wide lumen diameter relative to wall thickness / relatively large lumen / AW **;** tunica, externa / adventitia, as thick / thicker, than tunica media **;**	**2**

© UCLES 2020

Question	Answer	Marks
3(b)(i)	*any two from* tissue fluid and blood plasma do not have red blood cells **;** **A** blood contains red blood cells red blood cells are too large to pass through endothelial pores **;** *idea of* tissue fluid and blood plasma similar viscosity / blood more viscous **;** AVP **;** *ref. to* similar colour (versus blood is red)	**2**
3(b)(ii)	*any one from* taken up by / transported into / AW, (body) cells (from tissue fluid) **;** used by (body) cells to, synthesise polypeptides / proteins / enzymes **;**	**1**
3(c)	*any two from* contraction and relaxation **;** changes diameter of (lumen) of, trachea / bronchus / bronchiole **;** **A** (contraction causes) constriction control of air flow (in the bronchioles) **;** AVP **;** e.g. changed size of lumen during coughing / forced air out	**2**

Question	Answer	Marks
4(a)	converted the measured length (in mm) to μm (and dropped the, μm / units) *or* multiplied the measurement by 1000 **;**	**1**
4(b)	phosphate(-containing) heads are hydrophilic *or* triglycerides do not have a hydrophilic portion **;** lipid droplet is stored in / phosphate heads can interact with, cytosol / aqueous environment **/** watery environment **;**	**2**
4(c)	to, digest / break down, worn out organelles / waste metabolic products / toxic substances / AW **;** **A** invading pathogens	**1**

© UCLES 2020

Question	Answer	Marks
4(d)(i)	correct orientation of H and OH on, C1 <u>and</u> on C3 **;** OH added to carbon 6 **;** 	**2**
4(d)(ii)	positive result is coloured precipitate **;** *only allow if hydrolysis noted* **A** green / yellow / orange / brown / red, *for colour* *any two from* *ref. to* (heat with) Benedict's (reagent / solution) and, negative test / no colour change / remains blue, and test again with Benedict's **;** boil with (hydrochloric) acid **; A** hydrolyse with, acid / enzyme AVP **;** e.g. use a fresh sample to hydrolyse boil for 5-10 minutes with acid <u>cool</u> before neutralising neutralise (remaining acid) with alkali test with universal indicator paper (to check for pH7) compare with a control *allow one mark for heat with Benedict's and coloured precipitate*	**3**

Question	Answer	Marks
4(e)(i)	*any two from* *(higher K_m enzyme)* has a lower affinity for its substrate **;** **A** binds substrate less easily needs a higher concentration of substrate to reach, V_{max} / maximum activity / ½ V_{max} **;** less likely to be saturated with substrate **;** variations in substrate have greater effect on rate of reaction **;**	**2**
4(e)(ii)	*any two from* *in the vacuole **; A** in lysosome* qualified **;** e.g. has, acidic / low pH, environment contains, acid hydrolases / enzymes that require low pH non-regulatory trehalase needs acidic conditions for optimum activity cytosol has neutral pH so likely to be location of regulatory trehalase (and enzymes are in different locations) if in cytosol then this would mean low pH and other enzymes, would (partially) denature / work below optimum as acid conditions required, will be in area protected from rest of cell and vacuole has the tonoplast as barrier **A** lysosomes are membrane bound	**2**
4(e)(iii)	yes because they both work, within the cell / inside the cell **;**	**1**

Question	Answer	Marks
4(f)	*any three from* *likely to be regulatory trehalase / unlikely to be non-regulatory trehalase* nearer to optimum of regulatory trehalase (of S. cerevisiae) ; ora pH 6.5 / 6.6 nearer to pH 7.0 (than to pH 5.0 of non-regulatory trehalase) ; no / (very) low, activity at pH 4.5 *or* greater activity at pH 7 than pH 4.5(–5.0) ; *alternative* suggestion that enzyme could be a different form of trehalase ; because has different optimum pH to both regulatory and non-regulatory ; pH 6.5 / 6.6, rather than pH 7.0 or pH 5.0 ; *general* (so) likely to act in the cytosol with, neutral pH / pH7 ; unlikely to be found in vacuole / lysosome, with low / acid, pH ;	**3**
5(a)	*any two from* sequence of DNA nucleotides ; forms part of a DNA molecule / length of DNA / AW ; coding for a polypeptide ; **A** protein	**2**
5(b)	interphase ; **I** phases of interphase	**1**
5(c)(i)	two phosphates added to existing phosphate ; *base* = adenine ; *sugar* = ribose ;	**3**
5(c)(ii)	purine ;	**1**

© UCLES 2020

Question	Answer	Marks
5(d)	*any three from* to produce, many / AW, new blood cells **;** *idea that* large numbers are required to, transport oxygen / maintain immune system **;** produce genetically identical cells **;** loss of genetic material may not allow function / cells maintain function / cells are able to function **;** replacement of, old / damaged / dead, cells **;** AVP **;** e.g. to maintain healthy numbers of cells red blood cells have short life	**3**

© UCLES 2020

Question	Answer	Marks
6(a)	*any three from* *idea that* water moves down a water potential gradient / from a high to low water potential / AW **;** sucrose solutions produce differences in water potential inside the cell and externally *or* different concentrations of external sucrose solution produces different gradients of water potential **;** high concentration of, sucrose / solutes, is, lower / more negative, water potential **;** ora loss of water by osmosis out of potato cells lowers mass of block **;** ora for gain of mass no net gain or loss means water potential inside and out are equal **;**	**3**
6(b)	concentration where the, curve / line, crosses the *x*-axis (and use a reference table) *or* the concentration at which there is zero percentage change in mass (and use a reference table) **;**	**1**

© UCLES 2020